



Koneru Lakshmaiah Education Foundation

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OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 11

Team No.: 5

Project Title: **Kernel Monitor**

Team Members

Roll Number	Student Name	Signature
2520090094	Chaitanya	
2520090182	Bhardwaj	
2520089070	Salman	

1. Abstract

(Summarize the project idea, problem background, relevant Operating Systems/System Programming concepts, proposed Linux-based solution, and expected outcome. The abstract should not exceed 250 words.)

The **Kernel Monitor** is a Linux-based system programming project designed to monitor and display important information about the operating system and its running processes. The project addresses the need for a simple monitoring tool that helps users understand system resource utilization, process states, CPU usage, memory usage, and other kernel-level information. The project is closely related to Operating Systems concepts such as process management, CPU scheduling, memory management, system calls, process states, and resource monitoring.

The proposed solution will be implemented in **C/C++ on Linux** using POSIX APIs and Linux interfaces such as the `/proc` filesystem and system calls. The monitor will periodically collect system information and present it in a clear and readable format. It can display details such as process ID (PID), process state, CPU utilization, memory consumption, uptime, and overall system statistics.

The project provides practical experience with Linux system programming and demonstrates how user-level programs can interact with operating-system kernel information. The expected outcome is a lightweight and user-friendly **Kernel Monitor** capable of providing real-time or periodic information

about system and process activities.

2. Problem Statement

(Clearly describe the problem that the project intends to address and its relevance to Operating Systems and Systems Programming.)

Modern operating systems manage numerous processes and system resources simultaneously. Understanding CPU utilization, memory consumption, process states, and other system activities is important for monitoring system performance and identifying resource-intensive processes.

The **Kernel Monitor** project aims to develop a Linux-based monitoring application that collects and displays important system and process information in a simple interface. The project demonstrates how a user-space program can obtain information maintained by the Linux kernel through system calls and the /proc virtual filesystem.

This project is relevant to **Operating Systems and Systems Programming** because it involves process management, resource monitoring, system calls, Linux APIs, file handling, process identification, and interaction with kernel-provided information.

3. Objectives

(List the specific objectives of the project.)

1. _To develop a Linux-based **Kernel Monitor** using C/C++.
- 2.To monitor and display important system resources such as **CPU usage, memory usage, and system uptime**.
- 3.To retrieve and display information about running processes, including **PID, process state, CPU usage, and memory consumption**.
- 4.To understand and demonstrate Linux **system calls, POSIX APIs, and the /proc filesystem** used for accessing operating-system information.

4. Proposed Methodology

(Briefly explain how the project will be designed and implemented in Linux. Describe the major functionality, approach, and flow of the proposed solution.)

The Kernel Monitor will be designed and implemented as a user-space application running on Linux. The program will obtain system and process information from kernel-provided interfaces, primarily through the **/proc virtual filesystem** and relevant Linux/POSIX APIs.

The basic workflow will be:

Start → Collect System Information → Read Process Information → Calculate Resource Usage → Display Results → Refresh Periodically → Exit

The monitor will first collect general system information such as CPU statistics, memory availability, and system uptime. It will then scan the /proc directory to identify running processes. For each process, information such as PID, process state, CPU-related statistics, and memory usage can be obtained from files such as /proc/[PID]/stat and /proc/[PID]/status.

The collected information will be processed and displayed in a structured format. A periodic refresh mechanism can be implemented so that changes in CPU, memory, and process activity are visible continuously. Proper error handling will be included for processes that terminate while their information is being read.

5. Operating Systems Concepts / Linux APIs Used

(List the relevant Operating Systems concepts, Linux system calls, POSIX APIs, or mechanisms that will be used.)

OS Concept / Linux API / System Call	Purpose in the Project
Process Management	Identify and monitor running processes
Process ID (PID)	Uniquely identify individual processes
/proc Filesystem	Obtain kernel-provided process and system information
getpid()	Obtain the PID of the monitoring program
CPU Scheduling	Understand CPU utilization and process execution
Memory Management	Monitor memory consumption and availability

6. Individual Contribution

(Clearly specify the responsibility of each team member in the project.)

Roll Number	Student Name	Individual Responsibility
2520090182	Bhardwaj	Design overall Kernel Monitor and system architecture
2520090094	Chaitanya	implement process monitoring and /proc filesystem handling
2520089070	Salman	Testing, debugging, output formatting, and documentation

7. Tools / Platforms / Software Used

(List the programming languages, operating system, libraries, tools, and software planned for the project.)

Tool / Platform / Software	Purpose
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Linux / Ubuntu	Operating system and execution environment
C / C++	Implementation of the Kernel Monitor
Linux /proc Filesystem	Accessing kernel and process information
VS Code / Vim / Nano	Source-code development
Git / GitHub	Source-code management and collaboration

8. Expected Outcome

(Describe the expected functionality, output, and outcome of the project.)

The proposed **Kernel Monitor** is expected to provide a simple Linux-based interface for observing system and process activities. The application will display information such as:

- **CPU utilization**
- **Memory usage and available memory**
- **System uptime**
- **Running processes**
- **Process IDs (PIDs)**
- **Process states**
- **Process memory consumption**
- **Process CPU-related information**

The monitor will periodically refresh the displayed information, allowing users to observe changes in system resource utilization. The project will also provide practical understanding of **Linux kernel interfaces, process management, system calls, /proc filesystem, and system programming concepts**. Ultimately, the project will demonstrate how a user-space application can collect and present information maintained by the Linux operating system kernel.

9. GitHub Repository

GitHub Repository Link:  [suryakrishnabhardwaj-art/Kernel-Monitor](https://github.com/suryakrishnabhardwaj-art/Kernel-Monitor)

Repository Created: _____

☐ Yes ☐ No

Faculty Name and Signature

Faculty Name: Mrs_Alekhy Dasari_____

Signature: _____

Remarks: _____

Faculty Use

Project Approved: _____

☐ Yes ☐ No

Date: _____

Faculty Signature: _____